

The Rural Mobilization Gap in U.S. Place-Based Tax Credit

*Intermediary Selection versus Market Structure
in the New Markets Tax Credit*

IAN HELFRICH

Ph.D. Economics, Georgia Institute of Technology

ianthelfrich@gmail.com · github.com/ihelfrich/us-nmtc-viewer

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ABSTRACT

The U.S. New Markets Tax Credit (NMTC), enacted in 2000, subsidizes private investors with a 39% federal tax credit conditional on routing capital through certified Community Development Entities (CDEs) into projects in low-income census tracts. Over FY2001–FY2022 the program channeled \$66.6 billion of subsidized investment into 8,024 projects worth \$120.9 billion, so each dollar of subsidized investment sat alongside \$0.82 of other capital. Because the federal cost is the 39% credit on the equity basis and not the investment principal, that figure is a financing multiple, and the corresponding ratio per dollar of tax expenditure is roughly \$2.09. Non-metropolitan tracts received 19.6% of dollars, which sits at the program’s 20% administrative target, and yet project-level leverage in non-metro tracts is systematically lower (median 1.07× versus 1.19× metro). This paper decomposes the rural mobilization penalty using project-level fixed-effects regressions on the CDFI Fund’s public release. The naive estimate is a precise -0.262 rural penalty in mean leverage. Year, project-type, and state fixed effects shrink it to -0.172 . Replacing state effects with CDE effects moves the coefficient to -0.047 , and a fully nested model carrying both sets of effects gives -0.060 ; the mean regression is imprecise in either form (cluster-robust 95% CI $[-0.245, +0.152]$), while the median-regression analog is a precise -0.001 (SE 0.008), and within intermediary both the extensive margin (any private mobilization) and the intensive margin are null. An order-invariant Gelbach decomposition attributes -0.185 of the raw gap to CDE identity (CDE-cluster bootstrap 95% CI $[-0.347, -0.023]$; 86% of the explained movement), against -0.041 for project type. The aggregate rural leverage gap is therefore largely a between-CDE composition fact, not a within-CDE deployment penalty. A cross-CDE excess-mass test at the 20% non-metropolitan target finds no excess mass at the cumulative-deployment level, whether shares are measured in deal counts or in dollars and whether the sample covers the whole period or only the years after the proportionality rule entered the code in 2006 (dollar-share excess mass $\hat{B} = 0.0009$ after 2007; cluster-bootstrap 95% CI $[-0.013, +0.016]$). Compliance plausibly operates through allocation agreements rather than through deployment choices. The findings reframe the rural mobilization debate: the operative policy levers concern intermediary selection and capacity, not market-structure redesign.

JEL: H25 · H81 · R51 · G28

Keywords: New Markets Tax Credit · place-based policy · blended finance · mobilization ratio · Community Development Entities · rural finance

1 INTRODUCTION

The blended-finance literature has long debated which structures most effectively mobilize private capital alongside public concessional finance [Convergence Finance, 2024, Attridge and Engen, 2019, Lankes, 2021]. A persistent empirical limitation runs through that debate: the “mobilization ratio”—private dollars moved per public dollar committed—is rarely observed directly at the project level. Most evidence relies on survey-reported figures or on coarse program-level aggregates.

The U.S. New Markets Tax Credit offers a setting where the financing structure behind that ratio becomes visible at the deal level. For every project, the CDFI Fund’s public release records both the subsidized investment that the intermediary places (the Qualified Low-Income Community Investment, or QLICI) and the total project cost. I work throughout with their ratio, which measures how much total project capital accompanies each dollar of subsidized investment. Section 3 is careful about what this ratio is and is not: the government’s cost is the credit claimed against the equity basis, so the ratio describes financing structure rather than fiscal leverage, and I report both denominators wherever magnitudes carry weight. The panel covers a single instrument in a single country across twenty-two years and roughly 8,000 projects, which is an unusual amount of directly measured structure for this literature.

The question this paper asks is simple to state. Projects in non-metropolitan tracts carry less capital beyond the subsidized investment than projects in metropolitan ones. Is that gap a *market-structure* phenomenon—rural projects are fundamentally less leverageable—or an *intermediary-selection* phenomenon—the Community Development Entities that deploy in rural tracts are systematically weaker mobilizers everywhere they operate? The distinction carries direct policy content. A market-structure story argues for redesigning the credit; a selection story argues for reallocating it.

In this data the answer is composition, and the layers of the argument sit in plain view. The raw rural penalty in project-level leverage is -0.262 and precisely estimated. Layering in origination-year, project-type, and state fixed effects moves it to -0.172 . Introducing CDE fixed effects, which compare rural and urban deals *made by the same intermediary* and draw on the 163 of 343 CDEs that work both sides of the line, collapses the point estimate to -0.047 ; a fully nested model retaining state effects alongside CDE effects gives -0.060 . This paper is deliberate about what that null does and does not establish: the mean regression is imprecise (cluster-robust 95%

CI $[-0.245, +0.152]$), and on its own can reject only within-CDE penalties larger than 0.21. The precision lives elsewhere, and the paper reports it where it lives: the median within-CDE penalty is -0.001 (SE 0.008), rural deals within a CDE are no less likely to mobilize any private capital at all, and mobilize no less when they do. What differs across space is *which* intermediaries show up.

This paper sits inside two literatures. The NMTC evaluation literature has examined program targeting and neighborhood outcomes extensively [Gurley-Calvez et al., 2009, Freedman, 2012, Harger and Ross, 2016, Freedman and Kuhns, 2018, Theodos et al., 2020, 2022, Corinth et al., 2024], with program-evaluation foundations in Abravanel et al. [2013]; the place-based policy literature supplies the welfare frame [Glaeser and Gottlieb, 2008, Kline and Moretti, 2014, Neumark and Simpson, 2015, Diamond and McQuade, 2019]. Against both, this paper makes three contributions, since mobilization has been examined sparingly in either. First, and subject to the caveat that one cannot prove a negative about a literature this large, I am not aware of prior econometric work treating project-level financing structure as the central NMTC outcome, which brings the mobilization framework into a literature otherwise focused on neighborhood employment, demographic change, and price effects. Second, it implements a layered fixed-effects decomposition with CDE fixed effects, separating the within-CDE rural penalty from the between-CDE selection component—a decomposition that descriptive work by Theodos et al. [2021] motivates but does not estimate. Third, it brings the excess-mass logic of Chetty et al. [2011] and Kleven [2016] to the cross-CDE distribution of non-metro deployment shares around the program’s 20% target, a test that the target’s prominence in policy discussion would seem to invite and that I have not found in the published literature. No excess mass appears at the 20% line, in deal counts or in dollars, over the full period or over the years after the proportionality rule entered the code. Compliance plausibly operates through the allocation agreements that the public release does not report.

The rest of the paper proceeds as follows. Section 2 describes the program’s mechanics and the institutional heterogeneity of the CDE intermediary layer. Section 3 describes the public data release and the construction of the leverage outcome. Section 4 lays out the layered fixed-effects design, its interpretation, and its limits. Section 5 presents the decomposition, heterogeneity, robustness, and the bunching diagnostic. Section 6 draws out the policy reading, and Section 7 concludes. A causal companion design—a regression-discontinuity analysis at the low-income-community eligibility threshold—requires a tract-level ACS merge and is deferred to follow-on work.

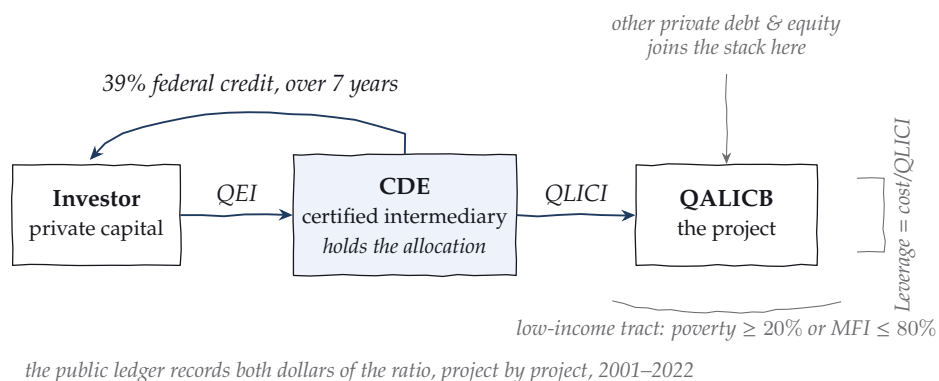


FIGURE 1. The program, as a working sketch. Private capital enters as a Qualified Equity Investment, the intermediary deploys it as a QLICI into a project in an eligible tract, and the 39% credit returns to the investor over seven years. The public ledger records both dollars of the leverage ratio for every project.

2 INSTITUTIONAL BACKGROUND

2.1 Program mechanics

The NMTC, enacted in the Community Renewal Tax Relief Act of 2000, awards investors a 39% federal tax credit, claimed over seven years, on Qualified Equity Investments (QEIs) made into certified Community Development Entities. CDEs in turn deploy that capital as Qualified Low-Income Community Investments into Qualified Active Low-Income Community Businesses (QALICBs) located in eligible census tracts. The public release distinguishes four QALICB types: real-estate projects (RE), non-real-estate operating businesses (NRE), special-purpose entities (SPE), and CDE-to-CDE loans. The credit's mechanics make the CDE the program's pivotal actor: it holds the allocation, originates the deals, and structures the capital stack around the subsidized tranche. Figure 1 draws the plumbing as it operates: capital walks left to right, the credit walks back along the top, and the CDE sits at the joint where every deployment decision is made.

2.2 Eligibility and the 20% non-metro minimum

Eligible low-income community (LIC) tracts are defined by a poverty rate of at least 20% or median family income at or below 80% of the area benchmark, with a targeted-populations override. Relevant here, a later amend-

ment directs the Treasury to write rules ensuring that non-metropolitan counties receive a proportional allocation of qualified equity investments. That instruction is IRC §45D(i)(6), added by the Tax Relief and Health Care Act of 2006 (P.L. 109–432, §102(b)), so it postdates the program’s creation in the Community Renewal Tax Relief Act of 2000 (P.L. 106–554, §121) and does not govern the earliest years of the sample. The statute names no percentage. The familiar 20% figure comes from the CDFI Fund’s administration of the requirement through allocation applications and agreements, and I treat it throughout as an administrative target and not a statutory threshold.¹ Aggregate deployment sits almost exactly at that line: non-metro tracts received 19.6% of QLICI dollars over the sample period.

2.3 *The intermediary layer*

The public release names 345 distinct CDEs in the transaction ledger (343 at the project level); 310 executed five or more transactions. The twenty largest account for 23.9% of QLICI dollars, so deployment is moderately concentrated, and specialization is extreme: among CDEs with at least five transactions, cumulative non-metro shares span the full range from 0% to 100%, with 39% of such CDEs never deploying non-metro and 6.5% deploying there at least four times out of five. This heterogeneity is the engine of the paper’s decomposition. If rural deals systematically flow through intermediaries who mobilize less *everywhere*, an aggregate rural penalty can arise without any within-intermediary rural disadvantage. The public release identifies CDEs by name; institutional-form classification (bank subsidiary, nonprofit CDFI, for-profit specialist) is a planned extension.

3 DATA

The analysis uses the CDFI Fund’s NMTC Public Data Release covering FY2003–FY2022 award years (released June 2024), which reports 19,907 QLICI transactions across 8,024 projects with origination years 2001–2022. Transactions are aggregated to the project level; the project is the unit at which total cost—and therefore leverage—is defined. Cleaning steps, field definitions, and file hashes are documented in the public repository’s data dictionary and provenance files, and the full pipeline from the raw work-

¹Effective dates and current contract language should be checked against primary sources before they are relied upon; Section 5.5 reports the mandate analysis for the full period and for origination years from 2007 onward.

TABLE 1. NMTC deployment by metro status, FY2001–FY2022.

	Metro	Non-metro
Projects	6,463	1,561
QLICI deployed (\$M)	53,566	13,045
Total project cost (\$M)	97,151	23,751
Mean leverage	1.99	1.73
Median leverage	1.19	1.07

book to every table in this paper runs from a single script sequence.

The outcome is project-level leverage,

$$\text{Leverage}_i = \frac{\text{ProjectCost}_i}{\text{QLICI}_i},$$

which I refer to as leverage throughout. Leverage is bounded below at 1 by construction, and the mass at 1 collects projects whose recorded cost equals the subsidized investment alone.

A word is owed on what this ratio measures, since the blended-finance framing invites a reading the data do not support. The QLICI is the investment the intermediary places, and it is not itself a federal outlay. The government’s cost is the 39% credit claimed against the qualified equity investment that funds the QLICI, spread across seven years, so a dollar of QLICI corresponds to roughly thirty-nine cents of tax expenditure under the substantially-all requirement that ties the two magnitudes together. The aggregates make the distinction concrete: \$120.9 billion of project cost against \$66.6 billion of QLICI leaves \$54.3 billion of other capital, which is \$0.82 for every dollar of subsidized investment and about \$2.09 for every dollar of implied tax expenditure. I therefore read leverage as a measure of financing structure. None of the paper’s identification turns on the choice of denominator, since a constant rescaling leaves the rural coefficient’s sign, its relative magnitude, and every test reported below unchanged. The baseline outcome is winsorized to [1, 20] to discipline a long right tail of large capital-stack deals; Section 5.4 shows the headline result is insensitive to the cap, including its removal.

Table 1 reports descriptives. Non-metro projects are 19.5% of the sample—again, the statutory line—and lag metro projects in both mean (1.73 versus 1.99) and median (1.07 versus 1.19) leverage. Figure 3 shows the shape of the gap: non-metro mass piles up near the 1.0 floor while the metro distribution carries a fatter right shoulder.

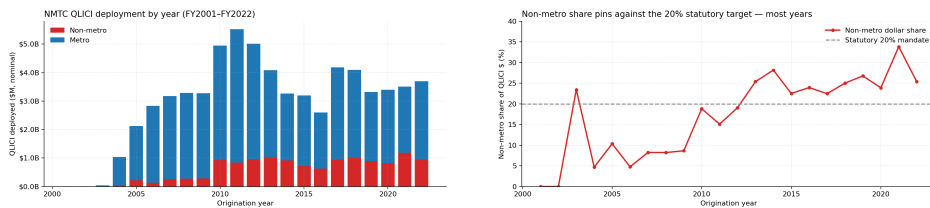


FIGURE 2. Deployment over time. Left: annual QLICI dollars, metro and non-metro. Right: the non-metro dollar share against the 20% statutory line.

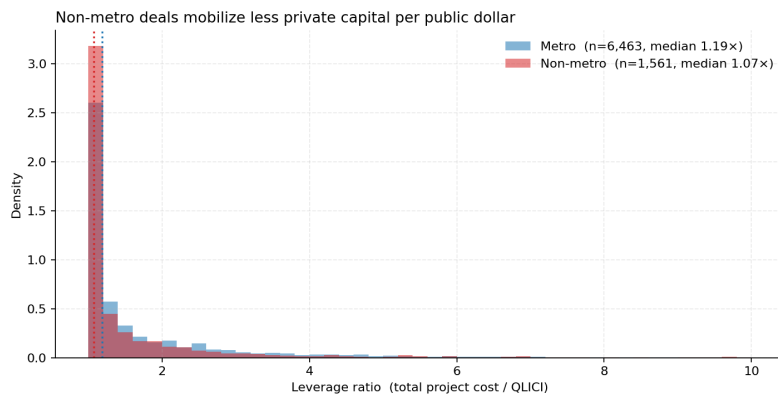


FIGURE 3. Project-level leverage distributions by metro status. Non-metro mass concentrates at the 1.0x floor (zero private mobilization); the metro distribution has the fatter right shoulder.

4 EMPIRICAL STRATEGY

4.1 Layered fixed-effects decomposition

Let R_i indicate a non-metro project. The paper estimates the sequence

$$L_i = \alpha + \beta R_i + \delta_{t(i)} + \eta_{q(i)} + \mu_{s(i)} + \gamma_{c(i)} + \varepsilon_i,$$

adding origination-year (δ), QALICB-type (η), state (μ), and CDE (γ) fixed effects in turn. One feature of the sequence deserves flagging at the outset, because it governs how the layered columns should be read. The workhorse CDE specification replaces the state effects of the preceding column rather than adding to them, since intermediary identity already absorbs much of what state effects capture. Column 7 of Table 2 therefore reports the fully nested model carrying both sets of effects, and I refer to it whenever a comparison across layers needs to be strictly nested. Each layer

removes a candidate composition story. The CDE step is the workhorse: with $\gamma_{c(i)}$ included, $\hat{\beta}$ is identified from rural–urban comparisons *within* intermediary, so it measures the within-CDE rural penalty—the market-structure component. The shrinkage of $\hat{\beta}$ upon adding CDE fixed effects measures the between-CDE selection component. Standard errors in CDE-fixed-effects specifications are clustered at the CDE level, the level of the fixed effect and the economic decision-maker.

4.2 Interpretation and limits

The decomposition is descriptive in the sense of selection-on-observables: within-CDE allocation of deals across rural and urban tracts is not random, and no LATE interpretation of $\hat{\beta}$ is claimed. What the design does establish is an accounting fact with economic content: whether the aggregate rural penalty survives comparisons inside the intermediary, where deal-structuring capacity, underwriting practice, and investor relationships are held fixed. A causal companion design—fuzzy RDD at the LIC eligibility threshold, run separately for metro and non-metro samples—requires tract-level ACS covariates and is deferred to the follow-on paper.

One further caution belongs here, because it constrains the language the paper is entitled to use. The between-CDE component is a composition fact; its “selection” reading is an interpretation. Rural market conditions could in principle cause low-leverage intermediaries to specialize in rural deployment, in which case part of the between-CDE component is market structure operating through intermediary business models. The paper’s secure claim is the within-CDE null; the decomposition locates the gap, and the discussion section is explicit about which readings of its location the data can and cannot separate.

4.3 Excess mass at the 20% target

For each CDE j , define its cumulative non-metro deployment share s_j , computed both as a count share, $n_{j,\text{nonmetro}}/n_{j,\text{total}}$, and as a dollar share of QLICI, since the requirement is written in terms of investment. If the 20% target binds at the deployment margin, optimizing CDEs should pile up at $s_j = 0.20$ and produce excess mass in the sense of [Chetty et al. \[2011\]](#) and [Kleven \[2016\]](#). The test compares empirical mass in a ± 2.5 percentage-point window around 20% against a degree-3 polynomial counterfactual fitted outside the window, over the 310 CDEs with at least five transactions, with a CDE-level bootstrap for inference. I avoid describing the target as a notch. The public data reveal no discontinuity in a CDE’s objective or

budget set at the threshold, and whatever consequence attaches to falling short operates through future allocation rounds that the release does not observe. Because the proportionality instruction dates from 2006, I report the test for the full period and again for origination years from 2007 onward.

5 RESULTS

5.1 *The decomposition*

Table 2 is the paper’s central exhibit. The raw rural gap (column 1) is -0.262 with a heteroskedasticity-robust standard error of 0.060 : non-metro projects mobilize about a quarter of a dollar less private capital per public dollar, and the difference is precise. Origination-year effects barely move it (column 2, -0.249): the gap is not a vintage story. QALICB-type effects (column 3) shrink it to -0.186 —rural deals skew toward operating-business types with structurally lower leverage, and that composition explains roughly a quarter of the raw gap. State effects (column 4) move it modestly further to -0.172 .

Introducing CDE fixed effects in column 5 collapses the rural coefficient to -0.047 with a CDE-clustered standard error of 0.101 , roughly a fifth of the raw gap in point-estimate terms, though the interval ($[-0.245, +0.152]$) is wide, and Section 5.2 quantifies exactly how much the mean regression can and cannot rule out. The median specification (column 6) is the precise one, placing the within-CDE rural penalty at -0.001 (SE 0.008). Because column 5 substitutes CDE effects for the state effects of column 4, column 7 reports the strictly nested model carrying year, type, state, and CDE effects together; it returns -0.060 (SE 0.099 , $p = 0.54$), so no part of the reading depends on that substitution.

Because sequential addition of fixed-effect blocks is order-dependent, Table 3 reports the Gelbach [2016] decomposition with the full model as reference, which is invariant to listing order. Of the -0.216 movement from the raw to the full specification, CDE identity contributes -0.185 (86% of the movement; CDE-cluster bootstrap 95% CI $[-0.347, -0.023]$), QALICB type -0.041 , and origination year a small offsetting $+0.010$. The share version of this statement—82% of the raw gap—carries a wide bootstrap interval ($[25\%, 255\%]$) because it is a ratio of estimates, and the paper therefore rests on the level claim: the CDE composition component is large, negative, and statistically distinct from zero, while the within-CDE component

TABLE 2. The rural leverage gap under layered fixed effects. Outcome: project leverage, winsorized [1, 20]. Columns (1)–(4) report HC1 standard errors; columns (5) and (7) cluster at the CDE level; column (6) reports the quantile-regression asymptotic standard error, which is not clustered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) Median	(7) OLS
Non-metro ($\hat{\beta}$)	-0.262*** (0.060)	-0.249*** (0.061)	-0.186*** (0.061)	-0.172** (0.067)	-0.047 (0.101)	-0.001 (0.008)	-0.060 (0.099)
Year FE	–	✓	✓	✓	✓	✓	✓
QALICB-type FE	–	–	✓	✓	✓	✓	✓
State FE	–	–	–	✓	–	–	✓
CDE FE	–	–	–	–	✓	✓	✓
R^2	0.002	0.015	0.022	0.039	0.131	–	0.141
N	8,024	8,024	8,024	8,024	8,024	8,024	8,024

is statistically indistinguishable from zero.

5.2 Precision, power, and margins

The mean and median estimates differ in precision by an order of magnitude, and the paper states plainly what each can support. For the mean specification, a one-sided 5% test can reject within-CDE rural penalties larger than 0.213—nearly the size of the raw gap—so the mean regression alone is weak evidence for a zero; the outcome’s long right tail, which inflates mean-regression variance, is the proximate reason. The median specification can reject penalties larger than 0.013. The paper’s null is therefore carried by the median estimate and by the margins, with the mean reported for comparability. (The median standard error is the quantile-regression asymptotic one; a CDE-clustered bootstrap for the median across 343 intermediary effects is computationally disproportionate for this draft, and it remains an outstanding item.)

Figure 4 shows the identifying variation itself, and it is the paper’s argument in a single drawing. Restricting to the 104 dual-market intermediaries with at least three projects on each side of the rural line, each vertical spine connects a CDE’s urban-book mean leverage to its rural-book mean. The spines climb a staircase from 1.0× to 4.0× across intermediaries, while the within-spine gaps distribute tightly around zero (median −0.12, interquartile range [−0.46, +0.36]): the range of levels across inter-

TABLE 3. Gelbach (order-invariant) decomposition of the raw rural gap, full-model reference. Contributions sum to $\hat{\beta}_{M0} - \hat{\beta}_{full}$ by identity (verified to 10^{-14} in the replication code).

	Contribution	Share of movement
Origination year	+0.010	-4.8%
QALICB type	-0.041	19.2%
CDE identity	-0.184	85.6%
Total ($\hat{\beta}_{M0} - \hat{\beta}_{full}$)	-0.216	100%
CDE component bootstrap 95% CI: [-0.347, -0.023] (499 CDE-cluster reps)		

mediaries runs several times wider than the spread of gaps within them.

The margins address a second way a mean could mislead: measured at a threshold of 1.001, 27.4% of projects record no capital beyond the subsidized investment, so a within-CDE penalty could in principle hide in the probability of drawing any additional capital at all. It does not. Within CDE, rural deals are 1.8 percentage points less likely to clear the floor (SE 1.8pp, $p = 0.34$), and among deals that mobilize, the rural coefficient is -0.039 ($p = 0.77$). Both margins are stable when the floor is moved to 1.05. Identification for all within-CDE results comes from the 163 of 343 CDEs that deploy on both sides of the rural line; these switchers originate 94% of all rural projects, so the comparison is broad, not a boutique subsample.

5.3 Heterogeneity by project type

Interacting the rural indicator with QALICB type (with CDE and year fixed effects retained) yields uniformly imprecise interactions: the rural main effect is 0.091 (SE 0.621), and no type-specific rural penalty is distinguishable from zero. The within-CDE null is not concentrated in, nor masked by, any single project type. The point estimates weakly echo the descriptive pattern—most negative for real estate, where private debt-stacking is most natural—but the data cannot reject homogeneity.

5.4 Robustness

Because a null this central should not depend on any single analytic choice, Table 4 probes the workhorse specification across alternative outcome transformations, winsorization caps, eras, samples, and clustering schemes. Unwinsorizing the outcome makes the naive gap *larger* (-0.338 , $p = 0.01$) and the within-CDE estimate remains null (-0.216 , $p = 0.48$): the winsoriza-

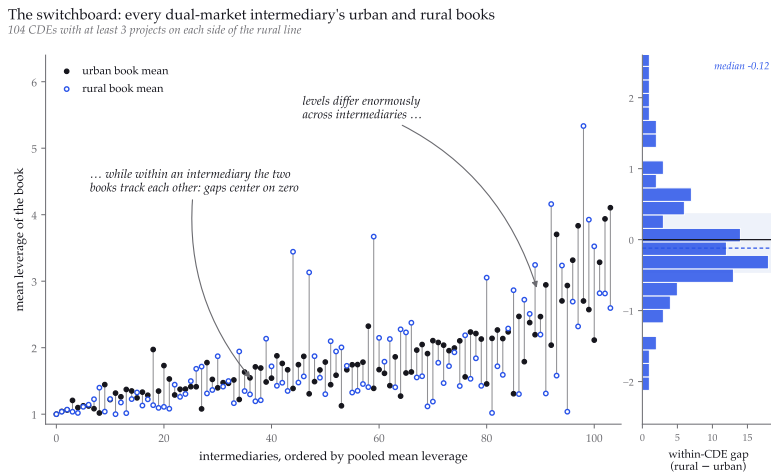


FIGURE 4. The switchboard. Every intermediary with at least three urban and three rural projects appears as a spine from its urban-book mean (filled) to its rural-book mean (open), ordered by pooled mean leverage; the marginal panel gives the distribution of within-CDE gaps against zero. The figure's sample rule and every displayed statistic are recorded in a machine-readable sidecar alongside the figure in the repository.

tion is not producing the headline result. A log outcome puts the within-CDE penalty at -3.2% ($p = 0.18$). Tightening or loosening the winsorization cap ($[1, 10]$, $[1, 50]$), splitting the sample at 2010, and restricting to single-CDE projects all leave the within-CDE rural coefficient statistically indistinguishable from zero. Restricting to the fifty largest CDEs, where the fixed effects are estimated most precisely, gives -0.100 (SE 0.148); two-way clustering by CDE and census tract moves the headline standard error from 0.101 to 0.105. The null is not an artifact of outcome trimming, era, stacked-allocation deals, subsample, or clustering choice.

5.5 No excess mass at the 20% target

Figure 5 plots the cross-CDE distribution of cumulative non-metro shares against the polynomial counterfactual. Excess mass in the window around the 20% statutory line is $\hat{B} = -0.0006$ —that is, *negative* and -2.2% of the counterfactual mass—with a CDE-level bootstrap 95% confidence interval of $[-0.014, +0.013]$; 47% of bootstrap draws are positive. There is no evidence that CDEs manage their cumulative deployment to the mandate line.

Two extensions guard against the objection that the test measures the wrong object. Measuring shares in QLICI dollars, which matches how the

TABLE 4. Robustness of the within-CDE rural coefficient. All rows except the first include year, QALICB-type, and CDE fixed effects with CDE-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	$\hat{\beta}$	(SE)	p	N
Unwinsorized outcome, no FE	-0.338**	(0.136)	0.01	8,024
Unwinsorized outcome	-0.215	(0.304)	0.48	8,024
Log outcome	-0.032	(0.024)	0.18	8,024
Winsorized at [1, 10]	-0.066	(0.073)	0.37	8,024
Winsorized at [1, 50]	-0.047	(0.138)	0.73	8,024
Origination 2001–2009	-0.044	(0.147)	0.77	2,488
Origination 2010–2022	-0.039	(0.107)	0.71	5,536
Single-CDE projects only	-0.088	(0.139)	0.53	6,366
Top-50 CDEs only	-0.100	(0.148)	0.50	4,229
Two-way cluster (CDE $\times$ tract)	-0.047	(0.105)	0.66	8,024
Bunching excess mass $\hat{B}$	-0.0006	95% CI [-0.0136, 0.0135], 999 CDE-bootstrap reps		

requirement is written, gives excess mass of +0.005 with a bootstrap interval of $[-0.008, +0.020]$. Restricting to origination years from 2007 onward, after the proportionality instruction enters the code, gives -0.003 in counts and $+0.001$ in dollars, with intervals of $[-0.016, +0.011]$ and $[-0.013, +0.016]$ across 291 intermediaries. The null holds in the units the mandate uses and in the era it governs.

Two readings are consistent with this null. First, the mandate may bind at the allocation-award level—commitments CDEs make in their allocation agreements—which the public release does not report; cumulative realized deployment would then sit wherever award-level compliance leaves it. Second, the cross-CDE distribution is close to bimodal, with urban specialists near zero and rural specialists far above 20%, leaving the mandate line in a low-density valley between modes. Both readings imply the same caution for policy discussion: the 20% line, prominent as it is in NMTC debate, is not where deployment decisions pile up.

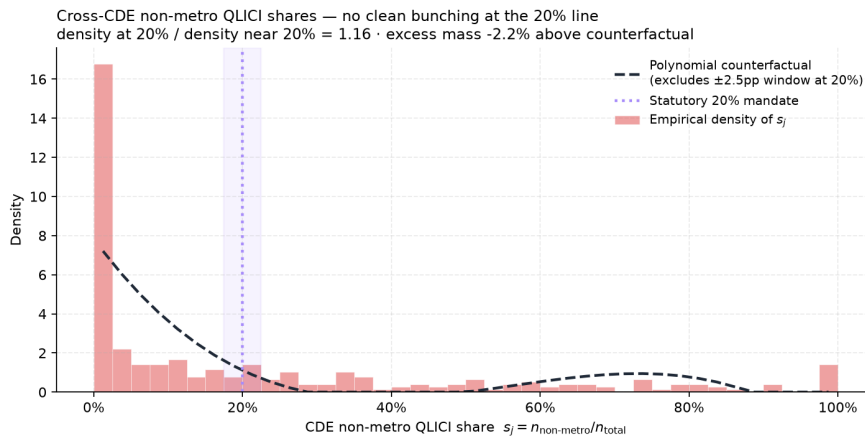


FIGURE 5. Cross-CDE distribution of cumulative non-metro deployment shares s_j , CDEs with at least five transactions ($n = 310$). The dashed line is a degree-3 polynomial counterfactual fitted excluding the shaded ± 2.5 pp window at the 20% statutory line.

6 DISCUSSION

6.1 What the results do and do not say

The results say that the within-CDE rural penalty is precisely zero at the median, null at both the extensive and intensive margins, and small but imprecisely estimated in the mean; and that the aggregate rural penalty—which is real—is located in between-CDE composition. They do not say that rural credit markets are no different from urban ones; they say the NMTC’s deployment data do not detect such a difference once the intermediary is held fixed. They also do not fully adjudicate *why* composition looks as it does: reading the between-CDE component as intermediary selection is the natural interpretation, but rural market conditions could in principle drive low-leverage intermediaries to specialize in rural deployment, which would put part of the between-CDE component back on the market-structure side of the ledger. Distinguishing those readings requires variation in CDE–market assignment that the public release does not contain. Nor, finally, do the results say that any particular CDE could not mobilize more in rural tracts; the estimates are averages over the fixed-effects distribution.

6.2 Policy reading

Reading the evidence this way carries three implications, in ascending order of specificity. First, the marginal rural deal does not appear constrained by market structure under the existing program design: when a given intermediary goes rural, its mobilization does not measurably suffer. Second, the aggregate rural gap is therefore amenable to *intermediary-allocation* levers—shifting allocation toward CDEs with demonstrated high-leverage capacity, independent of their current rural orientation—without redesigning the credit itself. Third, the 20% mandate as currently observed operates on a margin (cumulative deployment share) where behavior shows no bunching; if the policy goal is rural mobilization rather than rural deal counts, the mandate is pointed at the wrong margin.

6.3 Research program

This paper is the first phase of a sequence: the NMTC alone here; a multi-program U.S. comparison (NMTC, LIHTC, Opportunity Zones) next; and an international extension to development-bank project finance where the same mobilization framework applies. The LIC-eligibility regression discontinuity, which requires the ACS tract merge, is the immediate companion piece.

7 CONCLUSION

Directly observed project-level leverage in the NMTC shows a real aggregate rural mobilization penalty that is not visible inside the intermediary: the order-invariant decomposition places -0.185 of the raw -0.262 gap on CDE identity, the within-CDE penalty is precisely zero at the median and null at both margins, and the mean estimate, while imprecise, sits at roughly a fifth of the raw gap, and a strictly nested model carrying state and intermediary effects together returns -0.060 . The cross-CDE distribution of non-metro shares shows no bunching at the 20% statutory line. Subject to the interpretive caution that intermediary composition may itself respond to rural market conditions, the rural mobilization debate, at least for this program, is better posed in intermediary terms than in market-structure terms, and the operative levers are allocation levers.

A REPRODUCIBILITY

Every number in this draft regenerates from the public repository: `scripts/describe_nmtc.py` builds the analysis files from the raw CDFI workbook (hashes in `PROVENANCE.md`); `scripts/run_regressions.py` produces Table 2 and the bunching statistics; `scripts/run_robustness.py` produces Table 4; `scripts/make_paper_tables.py` writes the table fragments included here. The repository also serves a browser-based viewer with the full project map and specification explorer.

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